

ultra-low latency will also help in the growth of various new and emerging technologies such as the Internet of Things (IoT), artificial intelligence (AI), virtual reality (VR), augmented reality (AR), machine learning (ML), blockchain, and fixed wireless high-speed broadband network. Apart from leading the growth of new business operations and applications, it will also provide revenue platforms for telecom operators.

The countries that have successfully deployed 5G networks are Australia, South Korea, Taiwan, Japan, the US, UK, Germany, and a few others. They lead in terms of 5G network deployment, and site activations within the Asia Pacific have 5G non-standalone architectures that support 4G/LTE infrastructure for higher speeds.

As of now, India is home to a mere 322 million wireless broadband subscribers. However, as per a report by Edelweiss, the subscribers are expected to jump to one billion by 2025 with 5G. It is crucial as telecom is the very backbone of India's digital revolution.

Potential market areas of 5G communication

As per IDC's carrier transformation survey run at APeJ level, the top three 5G use cases for operator CXOs are massive IoT, remote-controlled automation, and multi-access edge computing. Under the current scenario, a gamut of interconnected products and services (smart home, smart refrigerators, smart HVAC, smart cars, CCTVs, smart devices, etc) might be an ideal play for operators, but India is still at a very nascent stage for 5G adoption.

Hence, in this context, let's find out what the experts have to say:

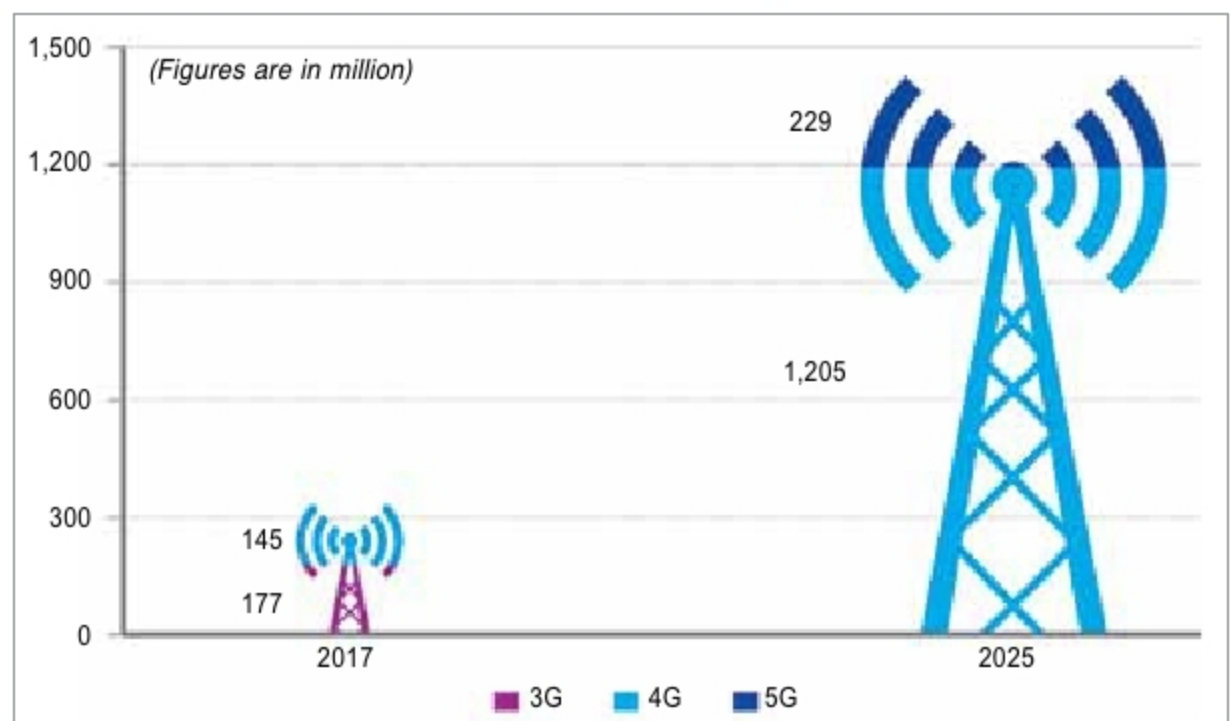


Fig. 1: Projected growth in wireless broadband subscriber base, post 5G rollout (Credit: Edelweiss research)

Faisal Kawoosa, founder and research analyst at techARC, says, "Due to the ongoing Covid-19 crisis, we have seen how much we can leverage from a robust and fast broadband connection. We believe that we have several use cases like industrial automation, high-speed home broadband, or immersive collaboration, be it for education, healthcare, or just office collaboration. Then there are use cases like ultra-rich multimedia content where we can use AR/VR for entertainment and gaming."

Rajan S Mathews, director general, COAI, says, "With 5G, technologies such as autonomous driverless cars, smart transportation and logistics, smart farming, drone patrolling for road safety, holograms, and remote robotic healthcare are expected to enter our world and become more and more pervasive by the day. The technology is poised to open up a plethora of possibilities in terms of business

models, better education, healthcare, smart cities, smart manufacturing, intelligent logistics, and overall enhanced lifestyles. The potential can be imagined by the fact that by 2026 there will be an estimated 619 billion dollars revenue potential for telecom operators."

Yash Jethani, research manager, Regional Telecommunications Team, IDC Asia/Pacific, says, "For operators, new areas are clearly in education, healthcare, and government (public sector as IDC defines it) in the Covid environment, but in the long term, driverless connected cars, 4K/8K streaming, smart cities, and mixed reality/immersive videos are some interesting use cases for Asia/Pacific operators. India's aggregate telecom spends on fixed data and fixed voice is thirty per cent and forty per cent, respectively, from B2C. B2C spend is even more predominant in the mobile data and mobile voice segments. Hence, for any future uptake in the market, the B2B focus also gives operators newer avenues to explore. With newer avenues, there will be numerous opportunities to capture under the government's push with Digital India."

When can we expect 5G in India

As per the Organisation for Economic Cooperation and Development (OECD) committee on digital economic policy, 5G will not only amplify the country's GDP but will also open up

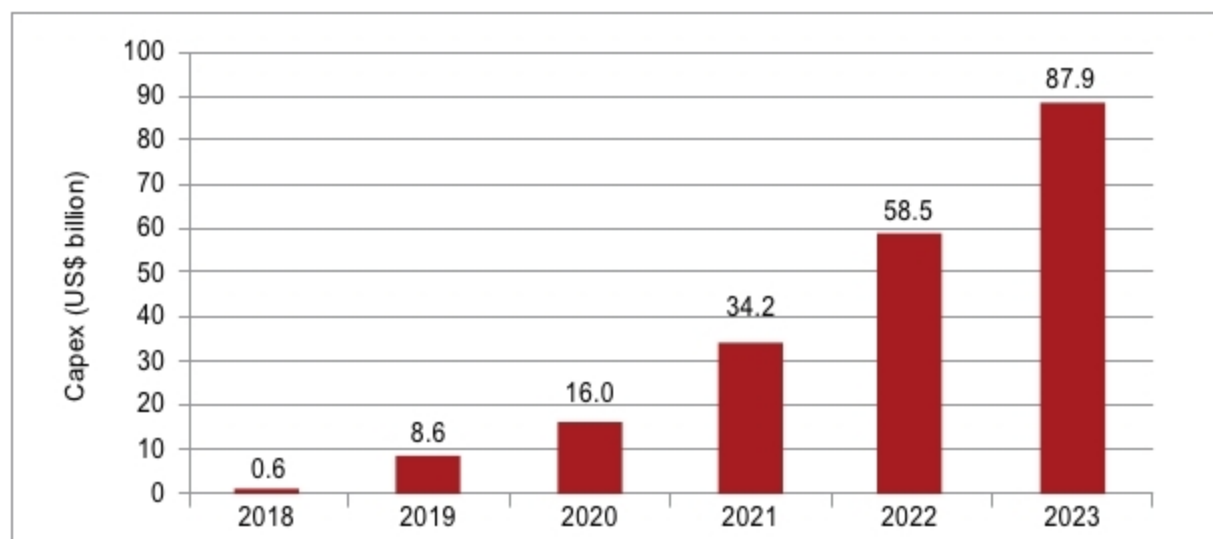


Fig. 2: Forecast of annual global 5G capex by mobile operators (Credit: Heavy Reading)